

YIMENG GU

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EDUCATION

Queen Mary University of London

Ph.D. in Computer Science

Advisor: Prof. Gareth Tyson

London, UK

Sept 2021 - Sept 2025 *expected*

Research Interests: Multimodal Learning, MLLM/LLM Applications, Fake News Detection

Carnegie Mellon University

M.S. in Electrical and Computer Engineering

Pittsburgh, PA, USA

Aug 2020 - Aug 2021

The Hong Kong University of Science and Technology

MSc. in Big Data Technology

Hong Kong, China

Sept 2019 - June 2020

Beihang University

B.E. in Electronics Information Engineering

Shenyuan Honors College (top 50/3000+)

Beijing, China

Sept 2014 - June 2018

RESEARCH EXPERIENCE

Queen Mary University of London | PhD Student

London, UK

Multi-MLLM Knowledge Distillation for Out-of-Context News Detection

Aug 2024 - May 2025

- Designed a news evidence-based chain-of-thought (CoT) prompt to obtain predicted labels and rationales from multiple 70B-parameter teacher MLLMs as knowledge
- Distilled teachers' reasoning capability to a 7B student MLLM via two stages: LoRA fine-tuning for rapid absorption of aligned knowledge, then DPO to improve discernment on conflicting knowledge
- Improved detection performance by 13.5% with 10x fewer parameters and only 10% labeled data

Contrastive Domain Adaptation for Out-of-Context News Detection

Sept 2023 - Apr 2024

- Pioneered to explore domain adaptation in out-of-context news detection
- Developed a domain-invariant feature learning approach incorporating contrastive learning and maximum mean discrepancy (MMD), reducing discrepancies between target and source domains
- Integrated confidence-based test-time training (TTT) for further target domain adaptation
- Improved domain adaptation performance by 2.6% on benchmark datasets

Detecting Multimodal Fake News with Gated Variational AutoEncoder

Mar 2022 - Oct 2022

- Designed perturbation experiments showing that multimodal news on social media contains substantial noise which undermines previous approaches' robustness
- Designed soft and RL-based gates to dynamically filter the noise in multimodal inputs
- Developed a gated multimodal VAE for robust news representation, achieving 3.7% performance improvement on benchmark datasets

WORK EXPERIENCE

Autodesk Research | Research Intern in AI Lab

London, UK

Generative AI for Identifying Mechanisms in Fusion360 Assemblies

Jun 2023 - Sept 2023

- Introduced a part-graph representation method by leveraging CLIP-based multi-view image encoding and TSP-guided ordering to transform part graphs into embedding sequences
- Proposed an AutoEncoder-latentGAN framework to first model the probabilistic distribution of a given part's neighboring parts in an assembly and then predict them
- Designed IoU-based and qualitative evaluation metrics; proposed method predicts neighbors of given parts effectively, improving IoU by 3% over baseline

- Annotated 360+ Chinese labor contracts to locate and extract key fields: *job position*, *job location*, *job description* and *part-time/full-time*
- Applied R-Net (reading comprehension) and BiLSTM-CRF (named entity recognition) to the task and achieved over 50% improvement in exact match (EM) and accuracy

ENGINEERING PROJECTS

The Design and Implementation of the Carcassonne Game

Mar 2021

- Designed system architecture and built a multiplayer Carcassonne game in Java (3.5k lines of code)
- Implemented interactive game interface with real-time tile placement and scoring using Java Swing
- Automated unit and integration testing with JUnit and Travis CI to validate core modules

SELECTED PUBLICATIONS

- Coarse-to-Refined: Multi-MLLM Knowledge Distillation for Out-of-Context News Detection
Yimeng Gu, Zhao Tong, Ignacio Castro, Shu Wu, Gareth Tyson
Under Submission, 2025
- Generate First, Then Sample: Enhancing Fake News Detection with LLM-Augmented Reinforced Sampling
Zhao Tong, **Yimeng Gu**, Huidong Liu, Qiang Liu, Shu Wu, Haichao Shi, Xiao-Yu Zhang
In *ACL*, 2025 (Oral Presentation)
- R²FND: Reinforced Rationale Learning for Fake News Detection with LLMs
Zhao Tong, **Yimeng Gu**, Qiang Liu, Haichao Shi, Shu Wu, Xiao-Yu Zhang
In *IJCNN*, 2025
- Contrastive Domain Adaptation with Test-Time Training for Out-of-Context News Detection
Yimeng Gu, Mengqi Zhang, Ignacio Castro, Shu Wu, Gareth Tyson
In *Pattern Recognition*, 2025
- GatedVAE: Detecting Multimodal Fake News with Gated Variational AutoEncoder
Yimeng Gu, Ignacio Castro, Gareth Tyson
In *ACM Web Science*, 2024
- MMVAE at SemEval-2022 Task 5: A Multi-modal Multi-task VAE on Misogynous Meme Detection
Yimeng Gu, Ignacio Castro, Gareth Tyson
In *NAACL SemEval Workshop*, 2022

HONORS AND AWARDS

- ACM WebSci'24 Travel Grant Apr 2024
- HKUST MSc (BDT) Excellent Student Scholarship 2019 (Top 7%) Apr 2020
- Excellent Undergraduate in Beihang University June 2018
- Second Prize in Science and Technology Competition in Beihang University (Top 15%) May 2017
- Honorable Mention in COMAP's Mathematical Contest in Modeling Apr 2016

LANGUAGE AND SKILLS

Computer/Technical	Python, Java, C/C++, Spark, SQL, git, gradle, IntelliJ IDEA, PyCharm, AWS, Azure, VS Code, Hadoop, Spark, AutoCAD, JUnit, Travis CI
Deep Learning	Pytorch, Tensorflow, HuggingFace, (M)LLMs, vLLM, LoRA FT, SFT, LMDeploy, Prompt Engineering
Language	English (fluent), Mandarin (native)